PROJECT PRESENTATION · POWER BI

Olist E-Commerce Analytics

From raw data to business insight with Power BI

Giorgi Lagvilava · Power BI Desktop, Power Query, DAX · Data: Olist public dataset (Kaggle), Jan 2016 – Dec 2018

R$13.5M Total revenue	**98.2K** Total orders	**R$137.4** Average order value
95.0K Unique customers	**4.09** Average review score	**91.9%** On-time delivery

All monetary values are in Brazilian reals (R$ = BRL).

Project overview

Olist is a Brazilian marketplace connecting small sellers with large online sales channels. The goal: turn its raw, multi-table dataset into an interactive Power BI report that a business user can explore without writing queries.

1. How do revenue and orders change over time?
2. Which categories and price bands drive revenue?
3. Which regions and states sell the most, and where is satisfaction lowest?
4. How concentrated is revenue among sellers?
5. How does delivery performance affect customer reviews?
6. How do customers pay, and when do they order?

Result: a three-page dashboard (Overview, Products & Sellers, Customers & Delivery) on a star-schema model, with slicers for Region, Year and Category.

Data and preparation

Olist public dataset, Kaggle (CC BY-NC-SA 4.0): real, anonymized orders across several CSV files

SCOPE OF THE FINAL MODEL

Period

Jan 2016 – Dec 2018

Orders · unique customers

98.2K · 95.0K

Active sellers

3.1K

Revenue (Brazilian reals, BRL)

R$13.5M

PREPARATION (POWER QUERY AND DAX)

Cleaned the source tables and set correct data types

Built dimension tables for customers, products, sellers, dates and states

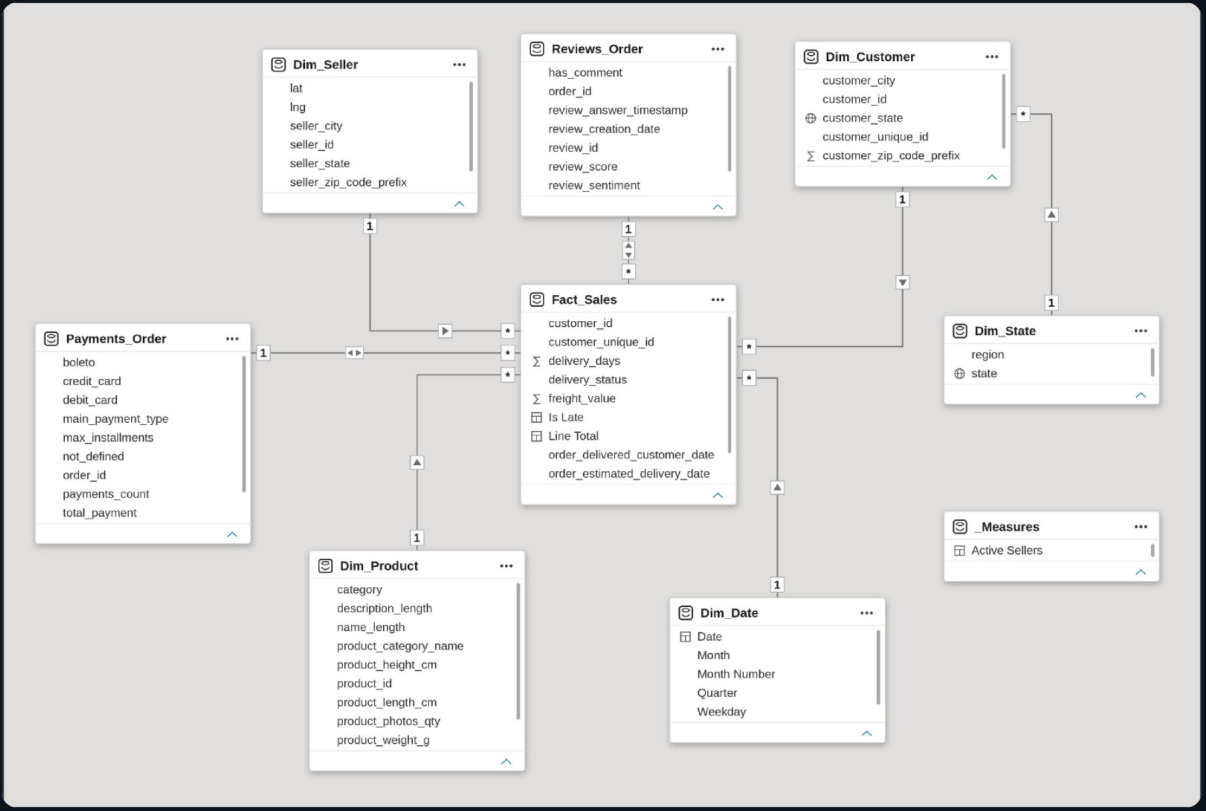
Created order-level tables for payments (one column per payment type, main type, installments) and reviews (score, comment flag, sentiment)

Added calculated columns for delivery time and status (delivery_days, Is Late) and line totals

Added a calendar table and a state-to-region mapping for time and regional analysis

Data model

Star schema in Power BI



One fact table, **Fact_Sales**, surrounded by dimension tables

Two order-level tables: **Payments_Order** and **Reviews_Order**

One-to-many relationships from each dimension to the fact table

Dim_State filters **Dim_Customer** for region-level analysis

All DAX measures live in a **_Measures** table

WHY A STAR SCHEMA?

Predictable filters, simple and fast measures, and the structure Power BI is designed for.

Dashboard · Overview

Page 1: the full picture in one view



R$13.5M revenue from **98.2K** orders; average order value R$137.4

Revenue grew steadily through 2017 and held near **R$1.0M per month** in Jan–Aug 2018

Credit card: **78.9%** of revenue; boleto 17.6%

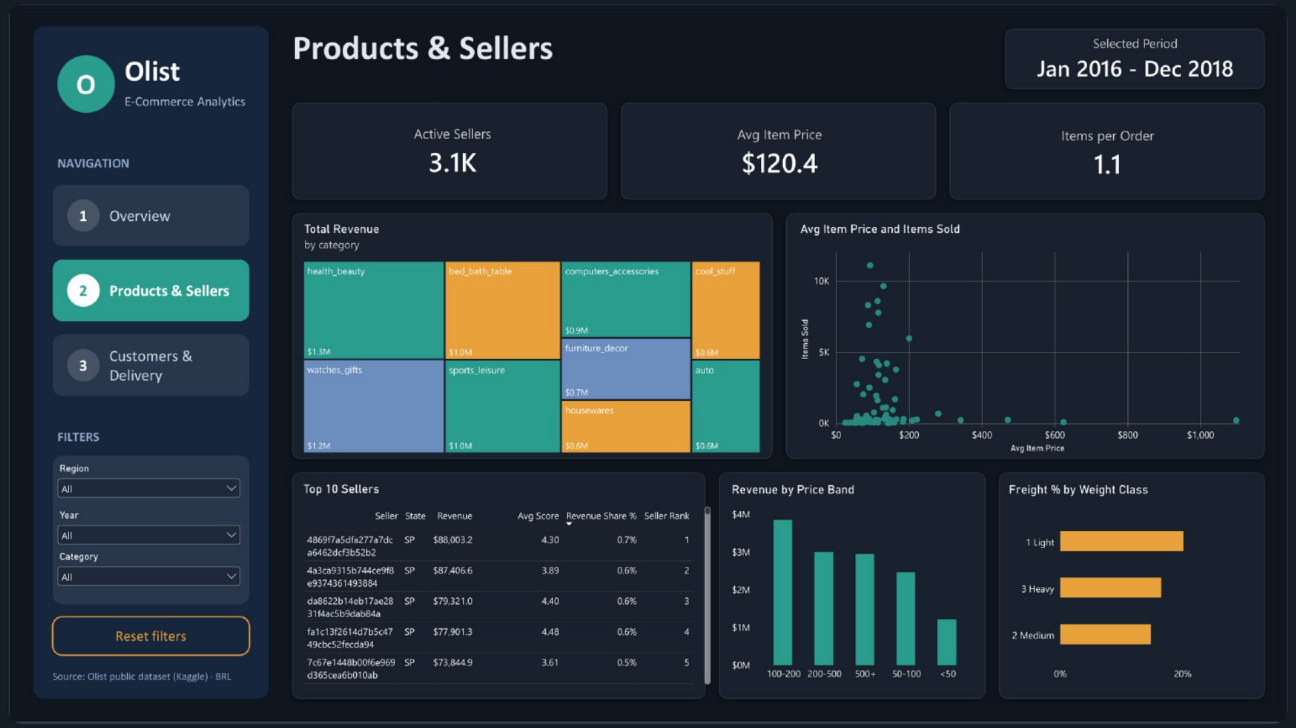
The **Southeast** generates **65.3%** of revenue and 68.6% of orders

READING THE 2018 LINE

The dataset ends in August 2018. The drop to zero afterwards reflects missing data, not a real decline.

Dashboard · Products & Sellers

Page 2: categories, prices and seller concentration



3.1K active sellers, average item price R$120.4, 1.1 items per order

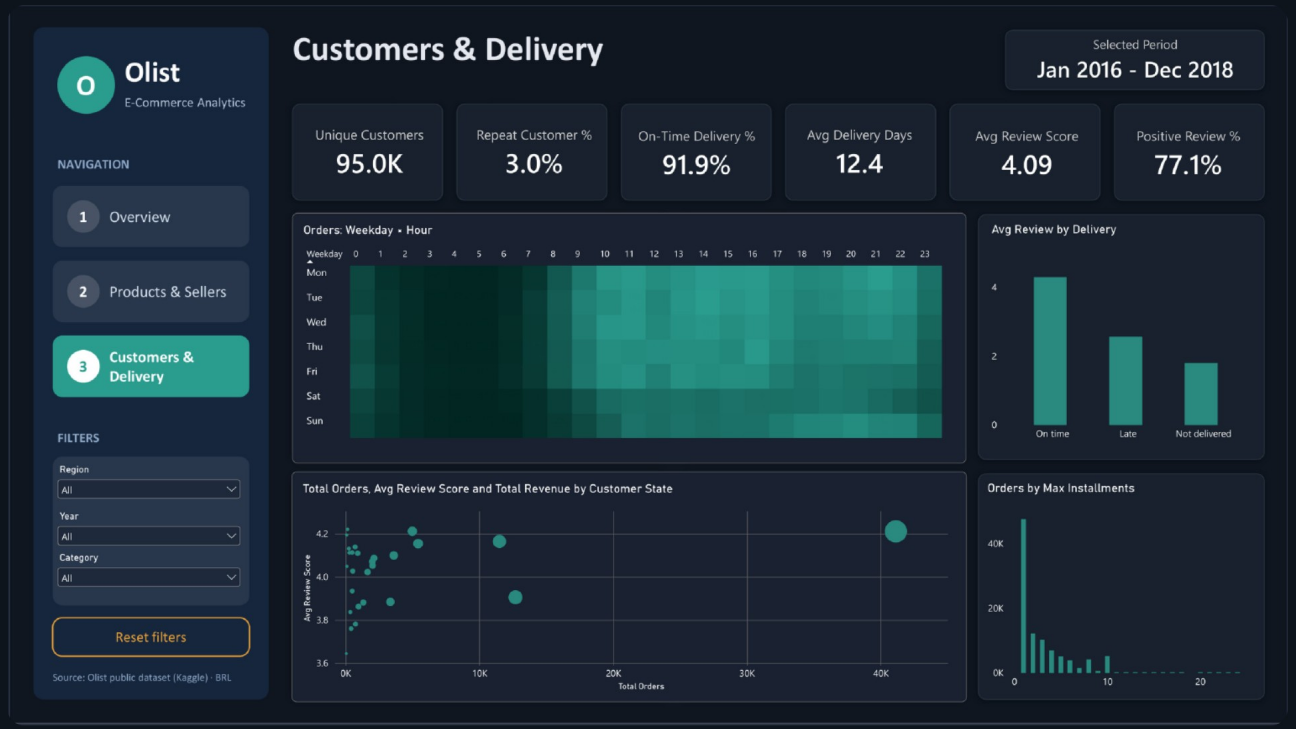
Top categories: health_beauty (R$1.3M), watches_gifts (R$1.2M), bed_bath_table and sports_leisure (R$1.0M each)

The **R$100–200 price band** earns the most revenue

No seller dominates: the top seller holds only **0.7%** of revenue

Dashboard · Customers & Delivery

Page 3: delivery performance and customer behavior



91.9% on-time delivery, 12.4 average delivery days, 4.09 average review score

On-time orders average about **4.3 stars** versus about **2.5** for late orders

Only **3.0%** of customers made a repeat purchase

Orders peak on weekdays during the day and in the evening; weekends are quieter

KEY INSIGHT

Delivery drives satisfaction

~4.3

average review score when the order arrives on time

~2.5

average review score when the order arrives late

About 1.8 stars lower for late orders. Only 91.9% of orders arrive on time, so reducing late deliveries is the most direct way to raise review scores.

Descriptive result from the dashboard; confirming causes would need further analysis.

Key findings and implications

What the dashboard tells a business user

Finding	Evidence	Implication
Sales concentrated in one region	Southeast: 65.3% of revenue, 68.6% of orders	Growth potential in other regions; concentration is also a risk
Delivery drives satisfaction	~4.3 stars on time vs ~2.5 when late	Cutting late deliveries is the most direct way to raise reviews
Satisfaction varies by region	Northeast has the lowest score (3.92)	Check whether delivery performance explains the gap
Very low repeat purchase	3.0% repeat customers	Retention and loyalty are a major untapped opportunity
Fragmented seller base	Top seller = 0.7% of revenue	Low dependence on any single seller

These implications interpret the dashboard's descriptive results; confirming causes would need further analysis.

Role of AI, limitations and next steps

Being transparent about method and scope

Role of AI

AI assistance (Claude) supported the project as a tool: guidance on dashboard layout, help with documentation and wording of this presentation.

The dataset choice, data model, measures and interpretation of results were reviewed and validated against the data in Power BI.

Limitations

- 2016–2018 only: 2016 has a few months and the end of 2018 is incomplete, which affects year-over-year comparisons
- The drop to zero after August 2018 most likely reflects the end of dataset coverage, not a real decline
- No cost or margin data, so profitability is not analyzed
- Reviews only reflect customers who chose to leave one

Next steps

- Delivery-time analysis by state to explain regional satisfaction gaps
- Cohort and retention analysis for repeat customers
- Forecasting for monthly revenue
- Publish the report to Power BI Service for sharing

Conclusion

The project covers the full analytics workflow: cleaning and modeling a multi-table dataset, then building an interactive dashboard that supports business decisions.

65.3%

Concentrated sales

of revenue comes from the Southeast region.

4.3 vs 2.5

Delivery drives satisfaction

average stars when on time versus when late.

3.0%

Retention gap

of customers made a repeat purchase.

Report, data model and documentation

github.com/giorgilagvilava2022/olist-ecommerce-powerbi-analysis

Thank you